
Exploring Generalization and Multi-Task Learning in Diffusion Policies with Language Conditioning

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1 Introduction

Diffusion models have recently emerged as a powerful class of generative models for robot policy learning. Chi et al. [1] demonstrated that formulating visuomotor control as denoising diffusion over action sequences, conditioned over observations, yields policies that are stable during training and capable of representing multimodal action distributions. However, a key limitation of this diffusion policy framework is that it requires training a separate model for each task, which limits scalability. A natural remedy is to condition the diffusion process additionally on language instructions, enabling a single policy to differentiate among tasks at inference time.

In this work, we re-implement the Diffusion Policy model¹, extend it with language conditioning to support multi-task learning, and conduct a series of ablation studies by training and evaluating on LIBERO [3], a comprehensive benchmark of 130 expert-demonstrated robotic manipulation tasks in simulation. We investigate three questions:

1) Do vision–language encoders trained in a shared embedding space improve multi-task policy learning?

Chi et al. [1] reported that pretrained vision encoders—including R3M [4], trained on large-scale egocentric video of humans performing manipulation tasks—underperformed compared to end-to-end training from scratch. However, their study did not include language conditioning and was limited to single-task settings. Liu et al. [3] tested a pretrained CLIP text encoder but did not pair it with a pretrained vision backbone. Neither work evaluated aligned vision–language encoders in the multi-task setting, leaving open the question of whether jointly pretrained visual and textual representations (e.g., CLIP or SigLIP) benefit multi-task policy learning.

2) Can paraphrased task descriptions encourage semantic grounding?

If a model only ever sees a single fixed instruction per task, it has no incentive to parse the instruction’s meaning; a constant string can function as a one-hot index that merely identifies the task. This behavior was observed by Liu et al. [3]: they compared conditioning on fixed natural-language task descriptions (e.g., “pull the top drawer open and put the bowl in it”) versus conditioning on a numeric task ID and found negligible difference in performance, suggesting a lack of semantic grounding. We hypothesize that augmenting training data with paraphrased instructions forces the language encoder to extract compositional semantics rather than memorize a fixed task identifier, which should improve generalization to unseen tasks. To evaluate this, we train models with and without paraphrased task descriptions, and test on both unseen tasks and seen tasks with different initial states.

3) How does fixed-capacity multi-task learning affect per-task performance?

Training a single model on many tasks with fixed capacity inevitably introduces competition for representational resources, and will perform worse than training the same model on a single task,

¹Our code repository is in [2]

but by how much is unclear. We test this by comparing single-task and multi-task models of nearly identical architecture and measuring the per-task performance gap.

2 Related Work

Diffusion Policy [1] adapts the DDPM diffusion model for robot visuomotor control, iteratively denoising random noise into action sequences conditioned on observations. This allows robot’s policy inherits several key properties necessary for robot action prediction such as temporal consistency and multimodal action distribution. They showed this approach outperforms prior methods (LSTM-GMM, IBC, BET) by an average of 46.9% across 12 simulation and real-world manipulation tasks. The model they created is the basis used for the experiments in this paper.

Jang et al [5] explores generalization of visuomotor policy to novel object-object or object-action pairs by conditioning a ResNet18 visual encoder using FiLM layers with language or video embeddings, and using different MLP heads that branches off the visual encoder as the control layer. They trained on 100 diverse manipulation tasks and showed 32% zero-shot success rate on 21 novel tasks specified via language. They find the bottleneck is likely the control layer rather than the task embedding, given similar training success rate between one-hot and language as task conditioning.

Hao et al [6] devise an approach to combine a VLM (Vision-Language Model) and diffusion model together. They use the VLM to create key-frames which the diffusion model uses to improve on its path. This demonstrates the effectiveness of combining a VLM with a diffusion model to improve the performance of the diffusion model and will be a crucial starting point if we are to integrate a VLM or VLA (Vision-Language-Action Model) as context into our diffusion model.

Liu et al [3] devise a system to create an unlimited amount of tasks. Additionally they created a high quality datasets of 130 different examples of them doing different tasks which will be the training for our diffusion policy. We plan to use the examples dataset they provide as a basis to train our diffusion model using imitation learning.

The evolution of vision-language pre-training has moved from the global Softmax-based contrastive learning used in CLIP [7] toward the more scalable pairwise Sigmoid loss introduced by SigLIP [8]. The most recent iteration, SigLIP2 [9], optimizes this approach for even larger-scale training, providing highly robust and aligned features across multiple languages. SigLIP2 is used to encode text descriptions and images received from the simulation. The reason for using it for both vision and language encoding is that since the two encoders are aligned, it is expected the diffusion model will have an easier time understanding the relation between how the description of the task aligns with the images.

3 Methods

Baseline. Our approach builds on the diffusion policy model introduced by Chi et al., which formulates a robot’s visuomotor policy as a conditional denoising diffusion process. Diffusion policy (Fig. 1) uses a 1D temporal convolutional UNet architecture that denoises an action sequence of length T_p (the prediction horizon). The action space can be either position-based (end-effector pose and gripper position) or velocity-based. We use position-based actions as they have been shown to be more effective. The denoising process follows DDPM [10] schedule. The model takes in past observation sequences of length T_o (the observation horizon) as conditioning inputs. These observations include proprioceptive state such as end-effector pose, image observations from one or more camera views, and the current diffusion timestep. All conditioning signals are concatenated channel-wise and injected into the UNet through FiLM layers [11], which learn channel-wise scale and shift vectors that apply an affine transformation at each stage of the UNet, providing global conditioning across the entire network.

The diffusion timestep is embedded by first converting to a sinusoidal positional encoding, which is then passed through a two-layer MLP. The proprioceptive state observations are included without any encoding: end-effector position is normalized to a $[-1, 1]$ range, and orientation is represented as a unit quaternion without further transformation. Image observations are encoded through a visual encoder such as ResNet-18, where each image in the observation history is independently passed through the encoder to produce a feature vector, all trained from scratch end-to-end with the policy.

During inference, only a short segment of the denoised action sequence is executed, of length T_a (the action execution horizon), before the model is rerun to produce a new prediction.

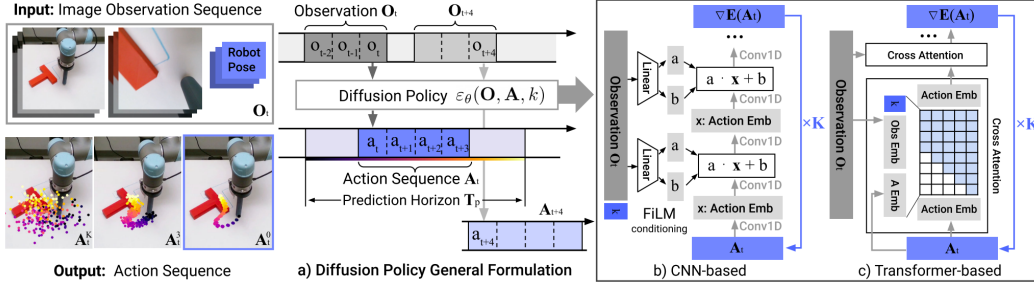


Figure 1: Overview of the diffusion policy architecture (figure reproduced from [1]). We use the CNN-based UNet variant for noise prediction.

Language Conditioning. As originally proposed, the diffusion policy has no notion of task identity and supports only a single task. We extend this by adding language conditioning to enable multi-task learning from a single policy. Specifically, we use a pretrained language encoder such as SigLIP or CLIP to encode a task description, and concatenate the resulting embedding with the other conditioning signals to be injected through the FiLM layers. To address the concern that the model may learn to use language merely as a task identifier without extracting semantic meaning, we add task description paraphrasing as a form of data augmentation: during training, we randomly sample from a set of alternative descriptions generated by an LLM rather than always using the canonical one.

Pretrained Visual and Text Encoders. To evaluate the effects of pretrained representations, including vision-language aligned encoders, we design the architecture to be modular with respect to its vision and language backbones, allowing different combinations of pretrained encoders such as ResNet, SigLIP, CLIP, and DINOv2.

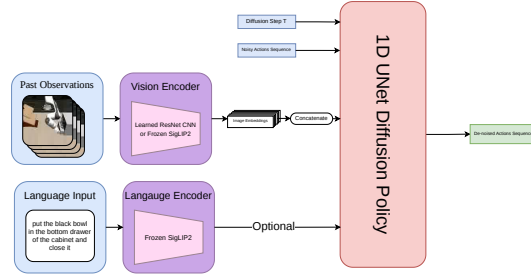


Figure 2: Diagram of the Overall Model Architecture. This demonstrates the inputs into each component of the model and the expected outputs.

4 Experiments

We conducted one experiment for each of the three research questions posed in Section 1. All models are trained and evaluated under the shared protocol described below.

Training. All models are trained exclusively on LIBERO-10 Fig. 3, which comprises 10 tasks with 50 expert demonstrations each. The models are trained with a similar setup as Diffusion Policy: using demonstrations 0-44 for each task we train for 50 epochs with a batch size of 64, and save a checkpoint every 5 epochs for validation. We follow almost identical setup as Diffusion Policy, summarized in Table 1.

Hyperparameter	Value
Optimizer	AdamW
Learning rate	1×10^{-4}
Weight decay	1×10^{-6}
LR schedule	Cosine with 500 linear warmup steps
Noise scheduler	DDPM with Square Cosine Schedule
Diffusion steps (train & inference)	100
EMA	Yes
Observation horizon	2
Prediction horizon	16
Action execution horizon	8
Diffusion step embedding dim	128
U-Net channel dimensions	[512, 1024, 2048]
Language projection dimension	256

Table 1: Training hyperparameters shared across all experiments.

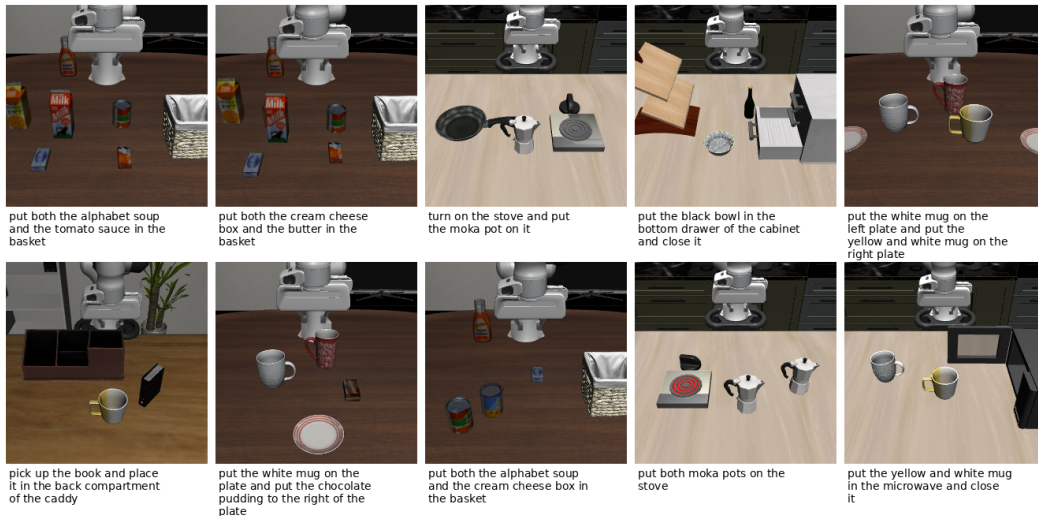


Figure 3: LIBERO-10 tasks: initial state and task description.

Evaluation protocol. LIBERO provides 50 fixed initial states per task to ensure reproducible evaluation. We partition these into a validation set (indices 0–19) and a test set (indices 20–39). Our evaluation metric is **success rate**: for each task and initial state, we simulate the policy to completion (up to 300 maximum steps) and record whether the task goal is achieved, provided by the simulation environment as a sparse reward. We report the per-task success rate averaged over initial states and the overall mean across tasks. During evaluation, we provide only the original task description. All evaluation results are recorded under `eval_results` folder in our repository.

Validation. After training, we use the validation split to select the best checkpoint for each model. We roll out each saved checkpoint on initial state indices 0–19 for every trained task and select the checkpoint with the highest mean success rate across tasks.

Testing. The selected checkpoint is tested on two sets of rollouts and we report two mean success rates. First, we evaluate on the 10 trained tasks using held-out initial states (indices 20–39), which measures in-distribution performance on unseen configurations. Second, we evaluate on 7 unseen tasks (Fig. 4) drawn from LIBERO-Goal, LIBERO-Object, and LIBERO-Spatial, using 20 initial states per task, which measures out-of-distribution generalization.

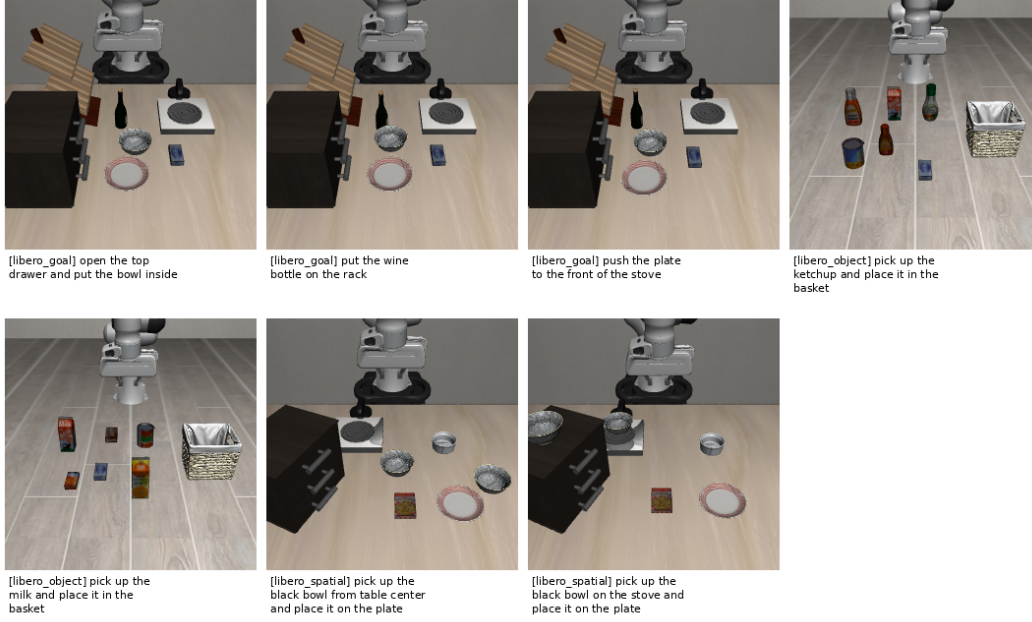


Figure 4: Unseen tasks used for generalization evaluation

4.1 Experiment 1. Effect of Pretrained Vision–Language Encoders

To study whether having a pretrained vision encoder and a pretrained language encoder in a shared embedding space improves multi-task policy learning, we compare two configurations, both trained with paraphrased task descriptions ²:

- **SigLIPv2 vision + text:** We use the siglip2-base-patch16-224 model from the HuggingFace Transformers library as both the vision and text encoder. Backbone weights are frozen and only the projection layers are unfrozen.
- **ResNet-18 vision + SigLIPv2 text:** ResNet-18 trained from scratch, paired with the same frozen SigLIPv2 text encoder.

By holding the text encoder and paraphrasing strategy constant, any performance difference is attributable to the vision encoder and its alignment with the language representation. We compare both on seen tasks with held-out initial states and on unseen tasks to measure performance.

The success rate for seen tasks on held-out initial states are shown in Fig. 5. The frozen pretrained vision encoder failed to complete any task, yielding a 0% mean success rate on seen tasks. In contrast, the ResNet-18 encoder trained from scratch achieved a 32.5% mean success rate. These results suggest that pretrained vision models must be finetuned for visuomotor policy learning; pretraining on large-scale web images alone provides no benefit, even when using pretrained multimodal models such as SigLIP.

The mean success rate on unseen tasks was 0% for both configurations. While there may be multiple contributing factors, this outcome is not unexpected given that unseen tasks involve novel goals and objects, and zero-shot generalization typically demands large-scale data and training. Without such pretraining, finetuning on these unseen tasks is likely necessary.

²Each Libero task comes with a single task description. We generate alternative descriptions using AI, roughly 40 for each task in Libero 10, and saved to a JSON file. See `diffusion_policy/dataset/task_descriptions.json` in our repository

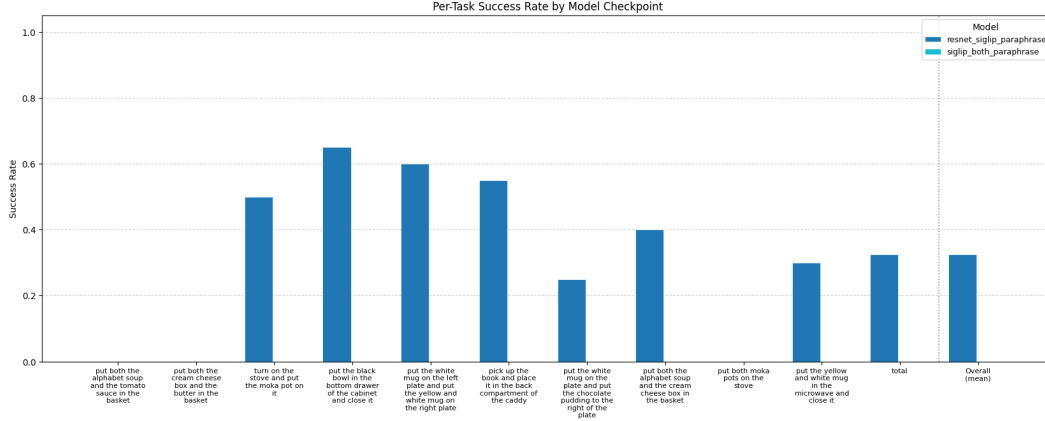


Figure 5: Experiment 1: Success rate in seen tasks (each task tested with 20 different initial states)

4.2 Experiment 2: Effect of Paraphrased Task Descriptions

To test whether paraphrasing encourages semantic grounding, we compare two models that share the same architecture (ResNet-18 vision encoder with a pretrained SigLIPv2 text encoder):

- **With paraphrase:** Each demonstration is paired with a randomly sampled paraphrase of its task description during training.
- **Without paraphrase:** Each task is always conditioned on its single canonical description.

We compared both models on seen tasks with held-out initial states, as shown in Fig. 6. The model trained with paraphrasing had 32.5% success rate, whereas the other have 7.5% success rate. However, it's worth to mention that the model trained without paraphrasing performed comparably on the two tasks it was able to solve, while achieving 0% success on the remaining 18 tasks. This suggests that task description paraphrasing enables the model to extract richer semantic information from task descriptions, allowing it to distinguish between different tasks more quickly and thus improving multi-task training efficiency. We expect that with longer training, the model without paraphrasing would likely learn to differentiate between tasks and achieve nonzero success rates on additional tasks as well.

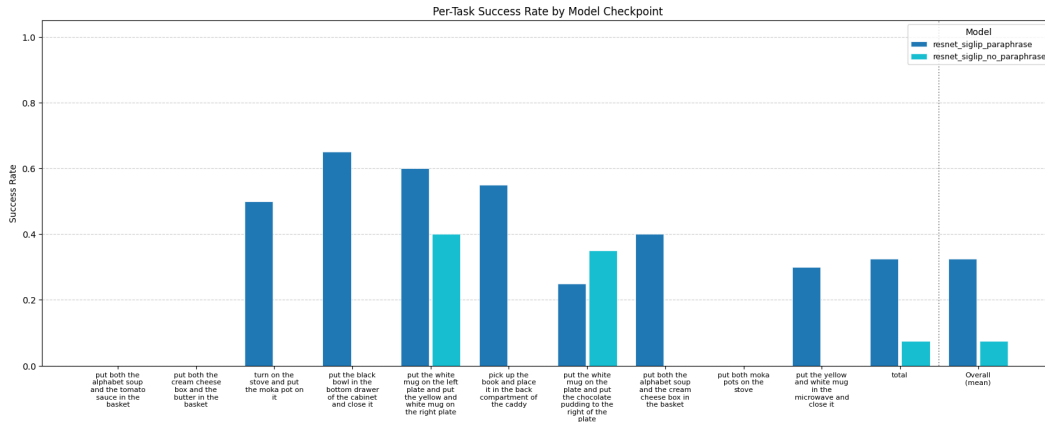


Figure 6: Experiment 2: Success rate in seen tasks (each task tested with 20 different initial states)

As in Experiment 1, both models didn't achieve any success on unseen tasks.

4.3 Experiment 3: Multi-Task vs. Single-Task Performance

To quantify the per-task cost of multi-task learning under fixed model capacity, we trained a single-task model on one randomly selected task from LIBERO-10: *"pick up the book and place it in the back compartment of the caddy"*. The model uses the same architecture as the best-performing multi-task model from Experiments 1 (ResNet vision encoder trained from scratch) but without language conditioning.

We trained it for 500 epochs to match the total training duration of the multi-task model, then compared its success rate against the multi-task model's performance on the same task. Ideally this analysis would cover more tasks, but time constraints limited us to one.

Surprisingly, the single-task policy (35% success rate) did not outperform the multi-task policy (55% success rate) despite being trained for the same duration (Fig. 7). One possible explanation is that multi-task training exposes the model to a wider distribution of robot arm trajectories across diverse tasks, resulting in more robust control. In contrast, a single-task policy, despite being trained with varied initial states, encounters only minimal variation in its trajectory distribution, which may cause it to overfit to a narrow region of the state space. Further investigation is needed to draw stronger conclusions.

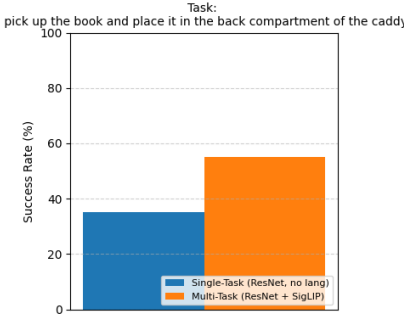


Figure 7: Success Rate on Libero 10 Task 3: Single-Task model vs Multi-Task model

One notable observation, also reported in LIBERO, is that lower noise prediction loss does not necessarily correlate with higher success rate. Fig. 8 shows the success rates for four checkpoints of the single-task model at different training epochs, used for validation and model selection. Despite achieving progressively lower training loss, later checkpoints performed worse in terms of success rate, suggesting possible overfitting. We did not evaluate all checkpoints due to the computational cost of evaluation, and instead selected the last four saved checkpoints for validation. However, this finding suggests that periodic success rate evaluation throughout training may be necessary to identify optimal checkpoints.

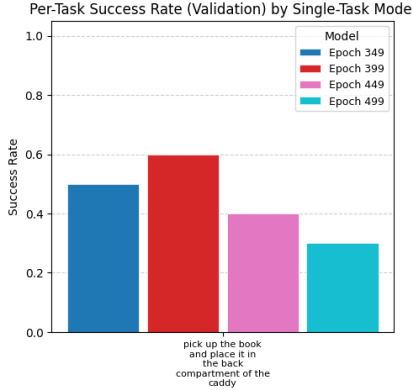


Figure 8: Success rate of the single-task policy on LIBERO-10 Task 3 across training checkpoints (Validation).

5 Conclusion

In this work, we investigated the integration of language conditioning into diffusion policies for multi-task robotic manipulation. While our models trained on the limited LIBERO-10 benchmark exhibited poor generalization to unseen tasks, achieving a 0% success rate on the unseen tasks, our experiments revealed several critical insights for multi-task policy learning. First, we demonstrated that frozen pretrained vision encoders—even those aligned with language, such as SigLIP2—are insufficient for visuomotor control without fine-tuning (0% success on seen tasks). In contrast, training a visual backbone from scratch yielded viable policies with an overall success rate of 32.5%. Second, we showed that augmenting training data with paraphrased task descriptions significantly enhances the model’s semantic grounding compared to using canonical descriptions, which only achieved a 7.5% success rate. This indicates that linguistic diversity forces the policy to extract compositional meaning from the embeddings rather than memorizing fixed identifiers.

Finally, our analysis demonstrated that multi-task training outperformed the model trained specifically on the task (achieving 55% vs. 35% success rate). This result is likely due to the regularizing effect of exposing the model to a broader, more diverse distribution of arm trajectories and seen situations. We also observed a clear decoupling between training noise prediction loss and task execution performance, underscoring the necessity of periodic environment testing to prevent overfitting during extended training.

In future work, we plan to transition our primary training to the LIBERO-90 dataset. Due to time and computational constraints, our current investigation was restricted to LIBERO-10. Because LIBERO-10 is fundamentally designed as an evaluation benchmark rather than a comprehensive training corpus, this restriction likely bottlenecked the model’s ability to generalize to unseen environments. Training on the standard LIBERO-90 dataset—which features a larger distribution of shorter-horizon tasks—will provide the appropriate foundation to determine if the model can effectively learn and generalize manipulation objectives. Furthermore, we aim to conduct deeper investigation into fine-tuning on various vision encoders and investigate the impact of visual augmentations to enhance the model’s overall robustness.

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